Jacobian Descent For Multi-Objective Optimization

Valérian Rey — Simplex Lab
valerian.rey@gmail.com
Joint work with Pierre Quinton

December 29, 2025

GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

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Theory

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

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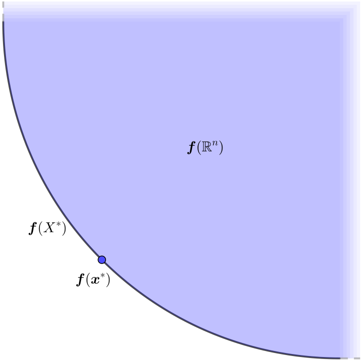

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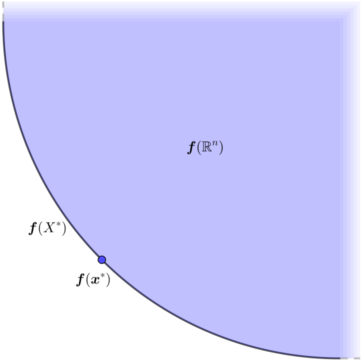

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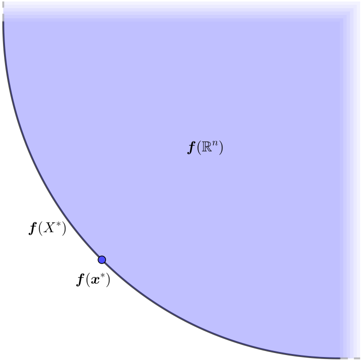
Pareto set: X^*
Pareto Front: $\boldsymbol{f}(X^*)$



First-order Taylor approximation

For any $\boldsymbol{x} \in \mathbb{R}^n$, the Jacobian of $\boldsymbol{f}$ at $\boldsymbol{x}$, is

$$\mathcal{J}\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} \nabla f_1(\boldsymbol{x})^\top \\ \nabla f_2(\boldsymbol{x})^\top \\ \vdots \\ \nabla f_m(\boldsymbol{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_1(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\boldsymbol{x}) \\ \frac{\partial}{\partial x_1} f_2(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_2(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\boldsymbol{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\boldsymbol{x}) & \frac{\partial}{\partial x_2} f_m(\boldsymbol{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\boldsymbol{x}) \end{bmatrix}$$

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

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$$\boldsymbol{f}(\boldsymbol{x} + \boldsymbol{y}) \approx \boldsymbol{f}(\boldsymbol{x}) + \mathcal{J}\boldsymbol{f}(\boldsymbol{x}) \cdot \boldsymbol{y}$$

We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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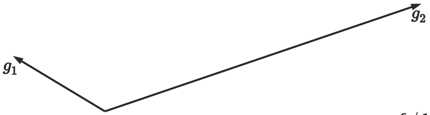
Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$
for $t \leftarrow 1$ **to** T **do**
    $\boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$
Output: $\boldsymbol{x}$

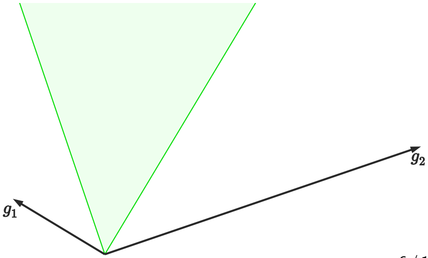
Unconflicting projection of gradients (UPGrad)

Given $J \in \mathbb{R}^{m \times n}$



Unconflicting projection of gradients (UPGrad)

Given $J \in \mathbb{R}^{m\times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\boldsymbol{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m\times n}$, $\mathcal{A}(J) \in C_J$.



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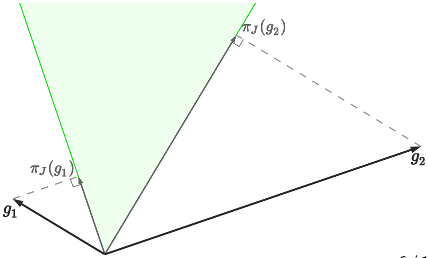
For any $\boldsymbol{x} \in \mathbb{R}^n$, the projection of $\boldsymbol{x}$ onto C_J is

$$\pi_J(\boldsymbol{x}) = \arg\min_{\boldsymbol{y} \in C_J} \|\boldsymbol{x} - \boldsymbol{y}\|^2$$



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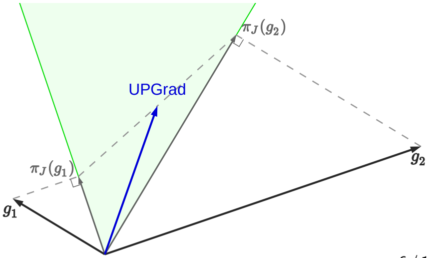
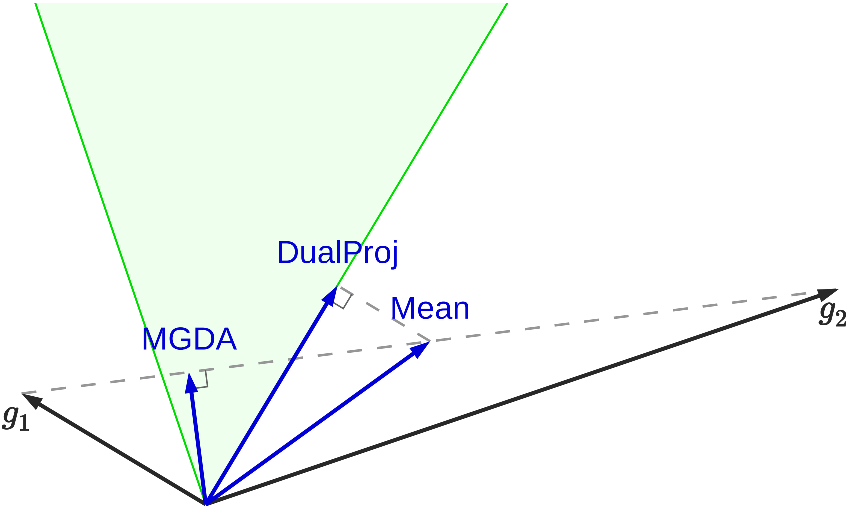
$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J\left(J^\top \boldsymbol{e}_i\right)$$



Other aggregators



Gramian-based JD

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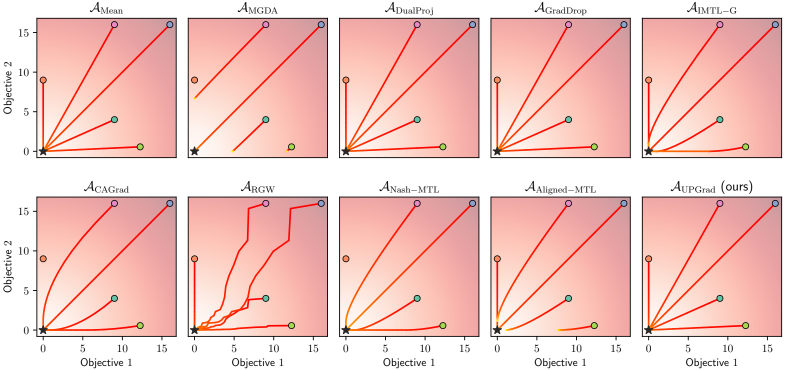
We have developed the **`autogram`** engine to compute $\mathcal{G}\boldsymbol{f}(\boldsymbol{x})$ efficiently.

This saves a lot of memory, so it's often much faster.

Applications and experimentation

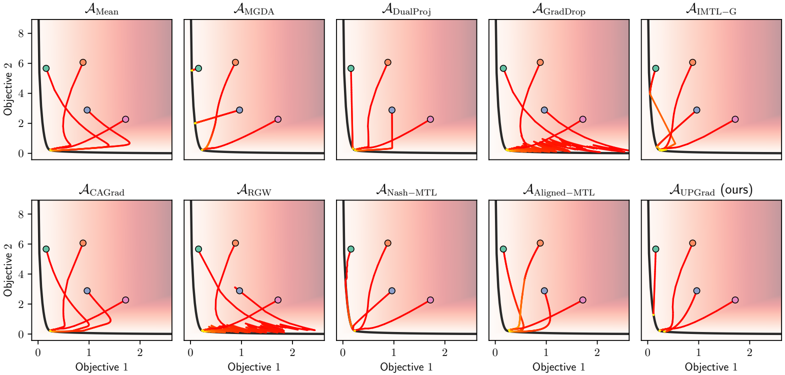
Synthetic example: element-wise quadratic function

Minimize $\boldsymbol{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.

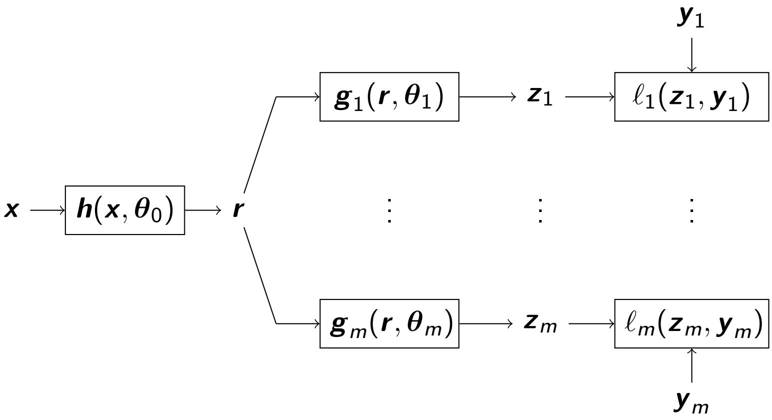


Synthetic example: convex quadratic form

Some convex quadratic function $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$, with highly conflicting gradients.



Multi-task learning



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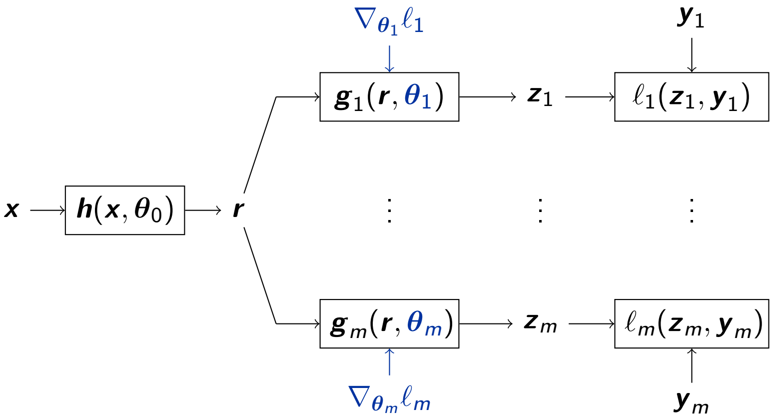

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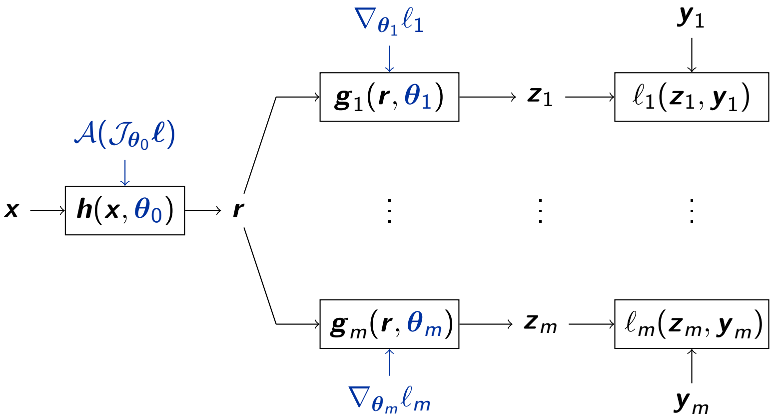

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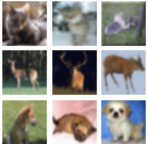
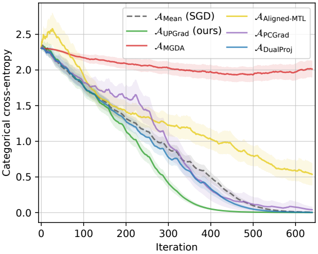
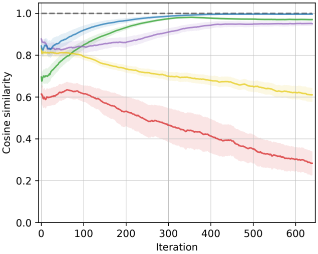
CIFAR-10



Training Loss



Update Similarity



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- Could JD (per-sample) help training diffusion models?
- Could JD (per-depth) help training recurrent models?
- Find new applications.

My research interests

Finding new ways to train neural networks.

Finding new architectures (that are hard to train with SGD).

Making algorithms faster (e.g. new custom tensors, etc.).

Contributing to the open-source community.